Code No: 183AK R22

Hall Ticket No:

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD

B.Tech II Year I Semester Examinations, February/March - 2025

OBJECT ORIENTED PROGRAMMING THROUGH JAVA

(Common to Computer Science & Engineering, Information Technology)

Time: 3 Hours Max. Marks: 60

Note: This question paper contains two parts A and B.
i. Part A is compulsory which carries 10 marks. All questions in Part A wish to be answered continuously.
ii. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART - A (10 × 1 = 10 Marks)

1.a) What is the difference between a class and an object? **[1]**

1.b) Which methods of deque enable it to be used as a stack? **[1]**

1.c) What is an abstract class? **[1]**

1.d) What is a package? How to define it and access it? **[1]**

1.e) How does Java support inter thread communication? **[1]**

1.f) List any four unchecked exception. **[1]**

1.g) What is adapter class? **[1]**

1.h) Give the AWT hierarchy. **[1]**

1.i) What is Swing in Java? How it differs from Applet. **[1]**

1.j) What are the differences between JToggle buttion and Radio buttion? **[1]**

PART - B (5 × 10 = 50 Marks)

UNIT - 1

2. Given two one dimensional arrays A and B which are sorted in ascending order. Write a program to merge them into a single sorted array C that contains every item from arrays A and B, in ascending order. **[10]**

OR

3. Explain the following with example a) this keyword. b) garbage collection. c) Constructor. **[10]**

UNIT - 2

4. What are the benefits of inheritance? Explain the various forms of inheritance with suitable code segments. **[10]**

OR

5. Write a program that generates random integers and stores them in a file named “rand.dat”. The program then reads the integers from file and displays on the screen. **[10]**

UNIT - 3

6. Write a program to read a f ile content and extract words using String Tokenizer class. Display the file if it contains the user query term/search key. **[10]**

OR

7. Describe the need of thread synchronization. How is it achieved in Java programming? Explain with a suitable program. **[10]**

UNIT - 4

8. a) What is the role of event listeners in event handling? List the Java event listeners. **[5]**

b) Write a program to implement the keyboard events in Java. **[5]**

OR

9. What is Layout manager? Discuss different types of layout managers used in JAVA with demonstration. **[10]**

UNIT - 5

10. Create a user interface using MVC architecture and swing concepts to book a seat in a bus [like redbus/abhibus applications]. **[10]**

OR

11. Explain the following: a) MVC architecture b) Anonymous inner classes c) Security issues in applets. **[10]**

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